

The Virtual Science Fair

The Virtual Science Fair was created as part of a community network system called MOOsburg (<http://linc.cs.vt.edu/moosburg/>). MOOsburg is a MOO—a collaborative environment that is modeled on a geographic space, in this case the town of Blacksburg. The goal of the Virtual Science Fair was to use MOOsburg tools (e.g., email, online chat, discussion forums, multimedia displays) to create a supplement to a traditional American science fair. In a science fair, students develop individual projects. They exhibit their projects at the fair, where the exhibits are judged and awarded prizes. Although a science project is developed over many months, the fair itself takes place over a few hours. Science fairs are open to the public, but normally are attended by the participating students, and their friends and families. The **high-level concept for the virtual science fair was to extend the boundaries of a traditional science fair**. By using MOOsburg, it was hoped that the fair could be made more attractive and accessible to a larger group of community members, and that activities of exhibit construction, viewing, and judging could be enriched.

Problem Scenarios

[1] Sally plans her exhibit on black holes.

Sally Harris is a high school sophomore who has been researching black holes for the past three months. This is a topic that has fascinated her for years and her biology teacher encouraged her to research it for the science fair this year, even though Sally won't be taking physics until next year. She has been in the science fair for the [past three years](#), so she knows a lot about the kind of projects students select, how they organize their exhibits, and what the judges look for.

She is a bit worried about the space and materials that is provided to everyone—a [standard 4x6 poster board](#), with two-foot shelf underneath for supporting [physical materials or models](#). This year she has explored some new methods, for example an Authorware simulation that illustrates her theory of black hole formation. But she knows from past years that there are [few electrical outlets in the gym](#), and she doesn't have a laptop to use in the exhibit anyway. She checks with the organizer Rachel Berris just in case, but Rachel confirms that the school district has no money for special resources like laptops, and that she will be able to use only battery-powered equipment. As she studies her simulation, Sally thinks of a way to turn the lack of computer support into a "feature": she will create a sequence of visualizations that can be flipped like a deck of cards to show the animation. In fact as she works, she gets into it and decides to create several variations, so that visitors can guess which one matches her project data and conclusions. She will then chart people's guesses as a dynamic element in her exhibit. She knows from experience that this is just the sort of thing judges will notice and [give her points for](#). Now she just has to figure out how to fit everything into the space she will have.

[2] Mr. King coaches Sally on her project.

Mr. King worked during his lunch hour so that he could save some time to come to the lab and work with Sally on her exhibit. He knows that

Sally has already done a lot of work researching black holes, but he wants to make sure she comes up with an effective layout, and that she covers all the standard project areas well. When she signed up, Sally told him she has done this [several times](#), so he figures the session will go pretty smoothly.

When he gets to the lab, Sally is already there and has begun to collect stacks of materials on a lab table, which happens to be the same 4x6 size as the bulletin board she will have at the fair. John can tell what is in the piles, but a quick look suggests that they hold materials related to the standard sections, Introduction at the top left, and so on. Sally shows Mr. King some photos of galaxies she has downloaded from the Web; she is very careful when handling these because they were printed in high resolution on [expensive glossy paper](#).

Mr. King notes that Sally already has the standard sections well-covered in her piles but thinks that she has too much content. Sally agrees and asks him to help her select the best pieces. He reminds her that both the judges and the visitors will be very influenced by first impressions as they walk up to the exhibit, so they first look for example materials that will look good from a distance. They use the table to lay these out, then choose supporting materials that provide the logical connections and explanations. Sally makes notes about figures and other documents that she still needs to create. She also talks to Mr. King about her idea for the interactive element, and he agrees that the [judges will love it](#). But he's worried that the exhibit will get too complex and messy, and suggests a decorated hanging container for visitors' experiments and comments.

[3] Alicia and Delia go to the science fair.

Alicia Sampson owns a hardware store. Her neighbor, Sally Harris, is participating in the science fair for the third year in a row, and Sally's mom has mentioned it several times. She wants to go to show support for Sally, but also wants to encourage her daughter Delia's interest in science. However, she also feels a little awkward about going—she knows the fair is in the [high school gym](#), and she has not been inside the school since she was a student there. Five years ago she even signed a petition asking the school board to tighten budgets in high school science. But she saw a science fair poster at Krogers today, and wonders if her husband can take Bobby to his basketball game. He's free, so she and Delia go on their own.

When Alicia and Delia walk into the gym, Alicia smiles, sharing briefly with Delia some pleasant memories of high school basketball games. After quickly scanning the open area, Alicia is amazed at the variety in projects. Some have just a few diagrams, others have complex physical models. She and Delia head for a project that has attracted a big crowd, but then she spies Marge (a former bridge partner) across the gym, so she and Delia go over to join Marge instead.

Marge is at Sally Harris' exhibit on black holes, and Sally is showing her a flip card animation of black hole formation. Alicia and Delia pause to listen, and Alicia is happy to see Delia [try out the animation](#). As they move on, Alicia is surprised to realize how much she has learned about black holes, a topic she's heard mentioned many times, but never really thought much about. Delia seems interested too; she tracks down one of her own friends and brings her back to meet Sally. Remembering Sally's poise and verbal skills, Alicia decides to follow up with her about a summer vacancy at her hardware store. As Alicia and Delia leave, they are talking about how they might get involved in next year's fair.

[4] Ralph judges the high school physics projects at the science fair.

For the last three years, Ralph—a retired civil engineer—has been a judge at the Blacksburg science fair, so he is not surprised to be asked again; he agrees because he knows that the organizer Rachel Berris counts on his experience on the judging team. In past years, he was occasionally able to get advance information about the exhibits, but this year could find out only the student names in advance. He is given the task of judging the high school physics projects; given prior years, he knows he will have to work quickly to get all five evaluated in the 90 minutes allotted. On the night of the fair, Ralph arrives promptly at 7pm, picks up his [forms](#) and begins the process of studying the exhibits and interviewing the students. His previous experience helps him to assign ratings in the specified categories, but as usual he finds it difficult, especially when trying to [compare across projects](#). At one point, he needs to evaluate two very nice projects in parallel, and finds himself running back and forth comparing details, annotating the scores with explanatory comments. Finally he is done; he signs the forms and submits them. Rachel and her assistant are already compiling a large stack of forms. Ralph is relieved to finish on time, and is eager to see how his evaluations will compare to those of the other judges.

[5] Rachel organizes a presentation for the superintendent.

Rachel was proud of the science fair projects that were exhibited this year, so she is delighted when Superintendent Mark Carlisle tells her he wants to show [highlights from the fair](#) at next month's school board meeting. She hopes that if he can impress the school board with some of the best exhibits, he will be able to make a case for increasing the resources available for next year's fair. Carlisle wasn't able to attend the fair himself, so Rachel briefly describes some highlights. She is careful to tell him about the problems with Sally's exhibit, where she was unable to demonstrate her black hole simulation because of lack of equipment and power sources.

Carlisle asks Rachel to collect sample materials for a 15-minute Powerpoint presentation. She spends a good part of the next week tracking down the [winning authors](#) and getting copies of their best poster elements. Some of the material is provided in digital form, but she has to use a scanner to digitize about half of it. After she has a good set of visuals, she browses them looking for the most impressive and self-explanatory examples, adding a few notes here and there to help Carlisle summarize the source projects. When she is done, she emails the Powerpoint presentation to Carlisle. She also decides to send him Sally's simulation file, just in case he has time to show it. Unfortunately, she later discovers that Carlisle's presentation backfires: the school board thought that the winning projects were of such high quality that there was no apparent need to increase resources for next year's event.

Situation Feature	Possible Pros (+) or Cons (-) of the Feature	Scenarios
Having a dedicated display board	⊕ leverages existing skills at grasping, moving, and constructing things in the world ⊕ encourages direct engagement and involvement by visitors ⊖ but each element consumes a fixed amount of physical space ⊖ but some pieces may be expensive or difficult to produce or copy	■ <u>Sally plans her exhibit on black holes.</u> ■ <u>Mr. King coaches Sally on her project.</u> ■ <u>Alicia and Delia go to the science fair.</u>
Repeated involvement by the same students	⊕ increases student familiarity and competence from year to year ⊕ encourages formation of a “science fair” community ⊖ but inexperienced students may find it hard to “break in”	■ <u>Sally plans her exhibit on black holes.</u> ■ <u>Mr. King coaches Sally on her project.</u>
Holding the fair in the high school gym	⊕ reinforces the feeling of community and school interaction ⊕ may bring back memories of other school events held in this place ⊖ but the organizers may need to coordinate with other school events (e.g., sports) ⊖ but the gym may not be set up with fair-specific resources (e.g., electricity)	■ <u>Sally plans her exhibit on black holes.</u> ■ <u>Alicia and Delia go to the science fair.</u>
Organizing the exhibits into poster boards	⊕ simplifies construction by providing a generic rectangular layout ⊕ provides a convenient visual overview for visitors as	■ <u>Sally plans her exhibit on black holes.</u>

	they approach [-] but some exhibit elements may be difficult to attach to such a board [-] but the uniform size may be too big or too small for some exhibits	
Competition among students for prizes	[+] rewards students who have contributed high quality time and effort [+] increases the excitement and engagement of event attendees [-] but students who try hard but don't win may be frustrated [-] but the diverse exhibits may be difficult to compare systematically	■ <u>Sally plans her exhibit on black holes.</u> ■ <u>Mr. King coaches Sally on her project.</u> ■ <u>Ralph judges the high school physics projects at the science fair.</u>
Judging forms or checklists	[+] are familiar and convenient objects for taking notes while walking around [+] organize and guide the judging procedure [-] but may be difficult to modify in exceptional cases [-] but will lead to considerable document processing if the fair is large	■ <u>Ralph judges the high school physics projects at the science fair.</u>
A summary of the winning projects	[+] gives explicit credit to the students who do the finest work [+] emphasizes that the event has an important outcome [-] but non-winners may feel left out or slighted if they are not mentioned [-] but such a summary may give a biased view of general project quality	■ <u>Rachel organizes a presentation for the superintendent.</u>